Image Enhancement

- Enhancements are used to make it easier for visual interpretation and understanding of imagery.
- The advantage of digital imagery is that it allows us to manipulate the digital pixel values in an image.
- Although radiometric corrections for illumination, atmospheric influences, and sensor characteristics may be done prior to distribution of data to the user, the image may still not be optimized for visual interpretation.
- Remote sensing devices, particularly those operated from satellite platforms, must be designed to cope with levels of target/background energy which are typical of all conditions likely to be encountered in routine use.
- With large variations in spectral response from a diverse range of targets (e.g. forest, deserts, snowfields, water, etc.) no generic radiometric correction could optimally account for and display the optimum brightness range and contrast for all targets.
- Thus, for each application and each image, a custom adjustment of the range and distribution of brightness values is usually necessary.



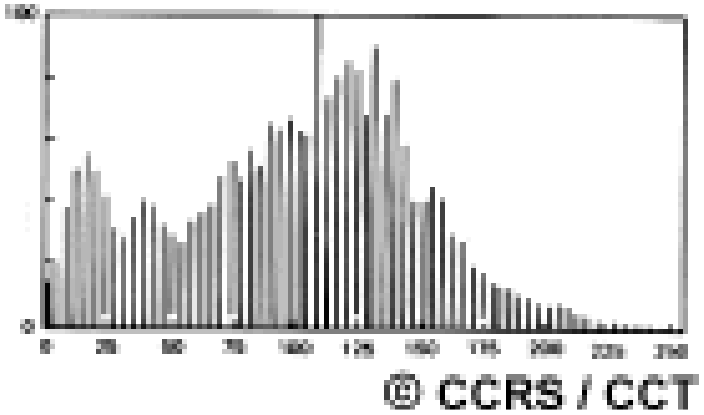
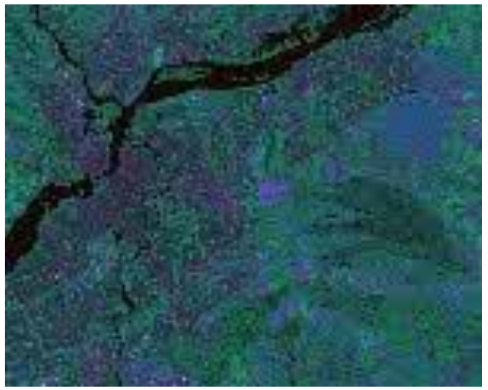
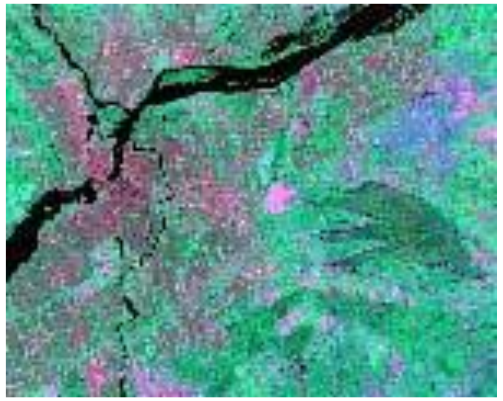


Image histogram

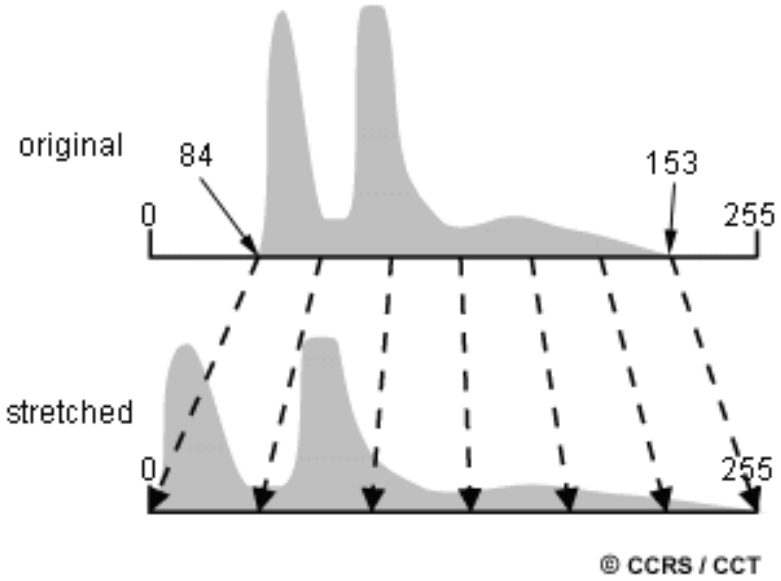
- In raw imagery, the useful data often populates only a small portion of the available range of digital values (commonly 8 bits or 256 levels).
- Contrast enhancement involves changing the original values so that more of the available range is used, thereby increasing the contrast between targets and their backgrounds.
- The key to understanding contrast enhancements is to understand the concept of an **image histogram**.
- ***A histogram is a graphical representation of the brightness values that comprise an image. The brightness values (i.e. 0-255) are displayed along the x-axis of the graph. The frequency of occurrence of each of these values in the image is shown on the y-axis.***

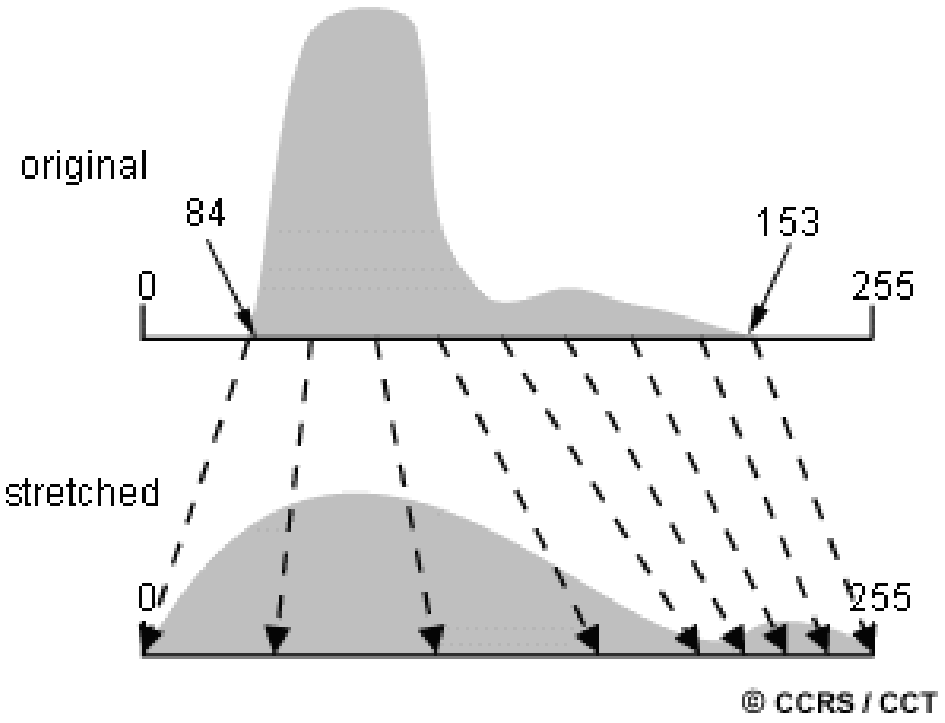




- By manipulating the range of digital values in an image, graphically represented by its histogram, we can apply various enhancements to the data. There are many different techniques and methods of enhancing contrast and detail in an image.

- The simplest type of enhancement is a **linear contrast stretch**. This involves identifying lower and upper bounds from the histogram (usually the minimum and maximum brightness values in the image) and applying a transformation to stretch this range to fill the full range.





- A uniform distribution of the input range of values across the full range may not always be an appropriate enhancement, particularly if the input range is not uniformly distributed.
- In this case, a **histogram-equalized stretch** may be better. This stretch assigns more display values (range) to the frequently occurring portions of the histogram.
- In this way, the detail in these areas will be better enhanced relative to those areas of the original histogram where values occur less frequently.
- In other cases, it may be desirable to enhance the contrast in only a specific portion of the histogram.

Spatial filtering

- Spatial filters are designed to highlight or suppress specific features in an image based on their **spatial frequency**.

- "Rough" textured areas of an image, where the changes in tone are abrupt over a small area, have high spatial frequencies, while "smooth" areas with little variation in tone over several pixels, have low spatial frequencies.

- A common **filtering procedure** involves moving a 'window' of a few pixels in dimension (e.g. 3x3, 5x5, etc.) over each pixel in the image, applying a mathematical calculation using the pixel values under that window, and replacing the central pixel with the new value.

- A **low-pass filter** is designed to emphasize larger, homogeneous areas of similar tone and reduce the smaller detail in an image. Thus, low-pass filters generally serve to smooth the appearance of an image. Average and median filters, often used for radar imagery are examples of low-pass filters.

- **High-pass filters** do the opposite and serve to sharpen the appearance of fine detail in an image.

- One implementation of a high-pass filter first applies a low-pass filter to an image and then subtracts the result from the original, leaving behind only the high spatial frequency information.

- **Directional, or edge detection filters** are designed to highlight linear features, such as roads or field boundaries.

- These filters can also be designed to enhance features which are oriented in specific directions. These filters are useful in applications such as geology, for the detection of linear geologic structures.

- ***Image transformations*** typically involve the manipulation of multiple bands of data, whether from a single multispectral image or from two or more images of the same area acquired at different times (i.e. multi-temporal image data). Either way, image transformations generate "new" images from two or more sources which highlight particular features or properties of interest, better than the original input images.

- Image division or **spectral ratioing** is one of the most common transforms applied to image data.

- Image ratioing serves to highlight subtle variations in the spectral responses of various surface covers.

- By ratioing the data from two different spectral bands, the resultant image enhances variations in the slopes of the spectral reflectance curves between the two different spectral ranges that may otherwise be masked by the pixel brightness variations in each of the bands.

- Another benefit of spectral ratioing is that, because we are looking at relative values (i.e. ratios) instead of absolute brightness values, variations in scene illumination as a result of topographic effects are reduced.

- Thus, although the absolute reflectances for forest covered slopes may vary depending on their orientation relative to the sun's illumination, the ratio of their reflectances between the two bands should always be very similar.